

Chapter 14

Agriculture — I

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

In Subsistence farming, the output is not very high and is consumed within the

1. village.
2. community.
3. family.
4. state.

Answer

family

Question 2

In Subsistence farming, there is a dominance of crops and is never any surplus for the

1. commercial, market
2. food, market
3. food, family
4. plantation, village

Answer

food, market

Question 3

The crops grown in Commercial farming are known as

1. Beverage crops

2. Cash crops
3. Fibre crops
4. Plantation crops

Answer

Cash crops

Question 4

Cultivation of crops and raising of animals is a feature of

1. Intensive farming
2. Extensive farming
3. Commercial farming
4. Plantation farming

Answer

Commercial farming

Question 5

Commercial farming is prevalent in areas where landholdings are and market economy is well developed.

1. large
2. small
3. consolidated
4. poor in fertility

Answer

large / consolidated

Question 6

..... agriculture is also known as 'slash and burn method'.

1. Subsistence
2. Shifting
3. Intensive
4. Extensive

Answer

Shifting

Question 7

..... and crops are associated with Shifting agriculture.

1. Rice and wheat
2. Tea and coffee
3. Yams and tapioca
4. Jute and cotton

Answer

Yams and tapioca

Question 8

..... is practised in many areas as an alternative to Shifting cultivation.

1. Rotation of crops
2. Intercropping
3. Nitrogen fixation
4. Dry farming

Answer

Intercropping

Question 9

Intercropping involves growing of two or more crops next to each other at the same

1. time
2. space
3. height
4. distance

Answer

time

Question 10

..... tree plantation is being promoted to check the practice of Shifting cultivation.

1. Mango
2. Orange
3. Apple
4. Timber

Answer

Timber

Question 11

..... type of farming involves cultivation of small farms using large inputs of labour, manures and fertilizers.

1. Extensive
2. Shifting
3. Intensive
4. Commercial

Answer

Intensive

Question 12

Which type of agriculture is mainly practised in the irrigated areas of the plains of northern India and the coastal area of south India?

1. Plantation
2. Shifting
3. Extensive
4. Intensive

Answer

Intensive

Question 13

The input of labour per unit area is low in farming.

1. Extensive

2. Intensive
3. Plantation
4. Subsistence

Answer

Extensive

Question 14

..... farming is an eco-friendly method as it involves less use of chemical fertilizers.

1. Subsistence
2. Commercial
3. Intensive
4. Extensive

Answer

Extensive

Question 15

A plantation is usually a over a large area which does not include naturally occurring plants.

1. monoculture
2. apiculture
3. horticulture
4. arboriculture

Answer

monoculture

Question 16

The farming which involves cultivation of single agricultural crop over large tracts of land or estates is

1. Subsistence agriculture
2. Commercial agriculture
3. Plantation farming
4. Mixed farming

Answer

Plantation farming

Question 17

Tea, coffee, rubber or spices are crops.

1. Plantation farming
2. Intensive farming
3. Extensive farming
4. Commercial farming

Answer

Plantation farming

Answer the following questions

Question 1(a)

What is meant by the term 'agriculture'?

Answer

Agriculture is defined as the cultivation of the soil in order to grow crops and rear livestock.

Question 1(b)

Why is agriculture said to be the backbone of the Indian economy?

Answer

Agriculture is said to be the backbone of the Indian economy as India is primarily an agricultural country as two-thirds of its population depends on agriculture. According to the Ministry of Statistics and Programme Implementation, agriculture and allied sector accounts for-

1. 16.5% of India's Gross Value Added
2. 17.32% of its Gross Domestic Produce
3. 13% share of total value of export
4. Employment to 58% of labour force either directly or indirectly

Question 1(c)

Mention any three problems of agriculture in India.

Answer

Three problems of agriculture in India are-

1. Indian agriculture is dependent to a large extent on the monsoons, which are uncertain, irregular and unequally distributed.
2. Old and inefficient methods and techniques of farming, inadequate irrigation facilities and inability of the farmers to purchase good quality seed and modern equipment because of paucity of funds and lack of latest know-how and inputs.
3. Soil erosion is not only a major cause for decreasing soil fertility but also results in loss of valuable crop land.

Question 1(d)

State three advantages of cooperative farming.

Answer

Three advantages of cooperative farming are-

1. It allows small farmers to pool together their resources and buy inputs at bulk rates and increase volume of products to open new markets.
2. It enables producers to negotiate for better prices, diffuse risks and share knowledge, skills and labour.
3. It improves the quality of life of member-farmers. It enables them to arrange time off, take care of children and get help in the form of extra hands when needed. Shared responsibilities, whole selling, producing or maintaining shared resources, lightens the work load of the farmers.

Question 2(a)

What is meant by Green Revolution?

Answer

Green revolution is regarded as the greatest revolution in the country which helped to transform the economy from food scarcity to food self-sufficiency.

It is a term used for describing the manifold increase in India's farm production and productivity, particularly in the case of major cereal crops like wheat consequent to the adoption of the 'New Agricultural Strategy' since the late-sixties.

Question 2(b)

State any two characteristics of Green Revolution.

Answer

Two characteristics of Green Revolution are-

1. Use of large capital and technological inputs
2. Adoption of modern scientific methods of farming

Question 2(c)

State any three negative impact of Green Revolution in India.

Answer

Three negative impact of Green Revolution in India are-

1. Land degradation due to overuse of fertilizers and pesticides
2. Drying aquifers
3. Vanishing biodiversity

Question 2(d)

What was the impact of Green Revolution on Indian agriculture?

Answer

Green Revolution had the following impact on Indian agriculture -

1. It enabled Indian agriculture to change from subsistence to commercial and market-oriented.
2. It led to the development of intensive agricultural production system that increased production and paved the way for self-sufficiency in respect of food grains.
3. The adoption of new technology under Green Revolution created more employment opportunities in agriculture sector.
4. It enabled the farmers to obtain increasing returns from agriculture by greater utilisation of agricultural inputs.
5. It increased rural prosperity.

Question 3(a)

Give two points of difference between Extensive and Intensive farming.

Answer

Extensive farming	Intensive farming
Farms are huge in size.	Farms are small in size.
It is highly capital intensive.	It is labour intensive system.

Question 3(b)

State any two advantages of Commercial farming.

Answer

Two advantages of commercial farming are-

1. It increases the yield rate of crops.
2. By utilizing high-end machinery for cultivating the lands, it takes less time to perform the agricultural operations.

Question 3(c)

Give a reason for each of the following:

- (i) Plantations are managed by large multinational companies.
- (ii) Fields are rotated instead of crops in shifting cultivation.
- (iii) In extensive agriculture, yield per hectare is low but total yield is large.

Answer

(i) Plantations are managed by large multinational companies because plantation crops are grown on large farms which are modern, scientific and self-contained units. Only one crop is grown on a large-scale. Enormous capital investment is required to set up a plantation and a large number of labourers are employed. Hence, only multinational companies can manage such large scale plantations.

(ii) Crops are not rotated in shifting agriculture because the people who practice shifting agriculture have become habitual of consuming specific crops. Hence, when the fertility of the current field is lost, they shift to another field instead of changing their crops.

(iii) In extensive agriculture, yield per hectare is low but total yield is large because it is practised in areas with large land holdings, using less labour and less chemical fertilizers and pesticides.

Question 3(d)

- (i) What type of farming is practised in areas where population is sparse and land is in plenty?
- (ii) Name two areas in India where such type of farming is practised.

Answer

- (i) Extensive farming is practised in areas where population is sparse and land is in plenty.
- (ii) Extensive farming is practised in Punjab and Haryana.

Question 4(a)

Give two points of difference between plantation and mixed farming.

Answer

Plantation farming	Mixed farming
Only one crop is cultivated using modern scientific methods.	Two or more crops are grown together.
Only crops are cultivated.	Crops and animals are raised simultaneously.

Question 4(b)

Give any two effects of globalisation on Indian agriculture.

Answer

Two effects of globalisation on Indian agriculture are-

1. Farmers have started adopting modern techniques of farming.
2. Establishment of food processing industries has increased employment.

Question 4(c)

Mention any three measures taken by the government to boost agricultural production.

Answer

Three measures taken by the government to boost agricultural production are-

1. Various land reforms have been introduced. Zamindari and all intermediaries have been completely abolished.

2. Creation of irrigation infrastructure and its optimum utilisation has been given greater importance.
3. The Government of India provides subsidy on fertilizers to ensure adequate availability of fertilizers to farmers at reasonable rates.

Question 4(d)

Agriculture in India is a gamble on the monsoon. Explain.

Answer

Indian agriculture is dependent to a large extent on the monsoons, which are uncertain, irregular and unequally distributed. Nearly 55% of the net sown area continues to depend on rainfall rather than irrigation. Hence, agriculture in India is a gamble on the monsoon.

Question 5(a)

How is shifting cultivation carried out?

Answer

In shifting cultivation, a patch of land is cleared by cutting and burning of the stumps. The ash is spread on the field as manure. After the land is cleared of trees, seed are sown in the ground and the land is cultivated. Neither ploughing of the soil nor any other agricultural practices are followed in this type of agriculture.

After 2-3 years, when the fertility of the soil is lost, the fields are abandoned and allowed to regain its fertility. The farmers move to a fresh piece of land and the same process is repeated.

Question 5(b)

Where is shifting cultivation practised in India? State any one disadvantage of shifting cultivation.

Answer

Shifting cultivation is practised in Assam, Kerala, Andhra Pradesh, Odisha, Himalayan region, Jharkhand and Madhya Pradesh.

The disadvantage of shifting cultivation is that it accelerates soil erosion and causes floods and silting in the lower reaches of the riverine flood plains.

Question 5(c)

State any two steps being taken by the government to discourage shifting cultivation.

Answer

Two steps being taken by the government to discourage shifting cultivation are-

1. The Government is promoting Intercropping as an alternative to shifting cultivation.
2. The Government is promoting large scale plantations of fast growing timber varieties to help in restoration of land and creation of wealth for the land owners.

Question 5(d)

What is subsistence farming? Mention two features of subsistence farming.

Answer

Subsistence farming is characterised by small and scattered land holdings and use of primitive tools, like hoe and digging sticks by family members. As the farmers are poor, they do not use fertilizers and high yielding variety of seeds in their fields.

Facilities like irrigation and electricity are generally not available to them. These factors result in low productivity. Most of the food production is consumed by the farmers and their families.

Two features of subsistence farming are-

1. Land holdings are small and scattered.
2. The farmers use traditional methods of agriculture.

Question 6(a)

What is Organic farming?

Answer

Organic farming is a unique production management system which promotes and enhances agro-ecosystem health, including biodiversity, biological cycles and soil biological activity, and this is accomplished by using on-farm agronomic, biological and mechanical methods in exclusion of all synthetic off-farm inputs.

Question 6(b)

What is Cooperative farming?

Answer

Cooperative farming refers to an organisation of farmers where farmers pool their resources in certain areas of activity. Each member farmer owns his land individually but carries out all the farming activities jointly with other farmer members. But profit is distributed in the ration of land owned by each member farmer.

Question 6(c)

What are the main characteristics of organic farming?

Answer

The main characteristics of organic farming are-

1. It protects the long term fertility of the soil by maintaining organic matter levels, and creating optimised conditions for biological activity within the soil.
2. It provides crop nutrients indirectly using relatively insoluble nutrient sources which are made available to the plant by the action of soil micro-organisms.
3. It maintains nitrogen self sufficiency through the use of legumes and biological nitrogen fixation as well as effective recycling of organic materials including crop residues and livestock manures.
4. It prevents weeds, diseases and pests by relying primarily on crop rotations, natural predators diversity, organic manuring, resistant varieties and limited thermal, biological and chemical intervention.
5. It provides attentive care and management of livestock, paying full regard to their evolutionary adaptations, behavioral needs and animal welfare issues with respect to nutrition, housing, health, breeding and rearing.

Question 6(d)

State why is organic farming gaining popularity in recent times.

Answer

Organic farming is gaining popularity in recent times because people have become aware of their benefits. They contain no chemical pesticides and fertilizers and are grown naturally with manure or compost and only natural pesticides and insecticides are used.

Children are more susceptible than adults to diseases caused by chemical pesticide residues in food and so parents prefer to give them organic foods. Another advantage is that organic foods do not contain growth hormones or antibiotic residues. Animals are often given growth hormones and antibiotics in animal feeds which are directly passed into animal foods like meat, poultry, fish, eggs, and dairy products and from foods to the consumers. In organic farming these practices are banned and animals feed outside in natural surroundings.

Thinking Skills

Question 1

Your penfriend, who lives in Australia, has read that India is an agricultural country. He wants to know the details about the nature of agriculture practised in India and the problems faced by the farmers. Write to your friend replying to his queries.

Answer

Agriculture plays a significant role in the Indian economy. The agriculture practiced here is diverse, including both subsistence farming, where crops are grown for personal consumption, and cash crop cultivation, which involves growing crops like cotton, sugarcane, tea, coffee, and spices for commercial purposes.

India underwent a Green Revolution in the 1960s and 1970s, which introduced high-yielding crop varieties, modern farming techniques, and increased use of fertilizers and irrigation. This revolution significantly improved agricultural productivity and helped the country achieve self-sufficiency in food production.

Indian farmers encounter several challenges, which impact their livelihoods and agricultural practices. These challenges include:

1. **Fragmented Land Holdings** — Land holdings in India are often small and fragmented, which makes it challenging for farmers to adopt modern farming techniques and achieve economies of scale.
2. **Dependence on Monsoons** — Agriculture in India heavily relies on monsoon rains for irrigation. Inadequate or irregular rainfall can lead to droughts, crop failure, and financial distress for farmers.
3. **Lack of Access to Credit** — Many farmers struggle to access formal credit, forcing them to rely on informal moneylenders who charge high interest rates. This perpetuates a cycle of debt and financial vulnerability.
4. **Inadequate Infrastructure** — Insufficient rural infrastructure, such as storage facilities, transportation networks, and market linkages, hampers farmers' ability to store, transport, and sell their produce efficiently.
5. **Climate Change Impact** — Indian farmers are increasingly facing the impacts of climate change, such as erratic weather patterns, changing rainfall patterns, and increased instances of pests and diseases.
6. **Old and Inefficient Techniques** — Most of the farmers in India use old and inefficient techniques of farming. Mechanisation is limited. Farmers continue to use traditional methods of irrigation.

Question 2

Your grandfather, a farmer, was a young boy when the Green Revolution began. What does he tell you about the pros and cons of the Green Revolution.

Answer

My grandfather experienced the Green Revolution as a young boy and he told me many things about the Green Revolution.

The advantages of Green Revolution as per my grandfather are:

1. **Increased Crop Yields** — The Green Revolution introduced high-yielding crop varieties and modern farming techniques, resulting in a significant increase in crop production and improved food security.
2. **Improved Livelihoods** — Higher crop yields meant increased incomes for farmers, improving their living standards.
3. **Food Self-Sufficiency** — The Green Revolution made the country self-sufficient in food production and ensured a stable food supply.

The disadvantages of Green Revolution as per my grandfather are:

1. **Environmental Impact** — Intensive use of chemical fertilizers and pesticides led to soil degradation, water pollution, and a loss of biodiversity.
2. **Income Inequality** — Large-scale farmers with better access to resources benefitted more, exacerbating income inequality and leaving small-scale farmers at a disadvantage.
3. **Dependency on External Inputs** — Farmers became reliant on costly inputs like fertilizers and pesticides.
4. **Loss of Traditional Knowledge** — The focus on high-yielding varieties led to a decline in traditional crop varieties, risking the loss of indigenous knowledge and agricultural diversity.

Question 3

Your uncle, who lives in Punjab, is a farmer involved in the cultivation of crops and livestock rearing. He sells the crops and dairy products in the market. Name the type of agriculture practised by him, the crops he grows and the main features of the agriculture practised by him. Why is this type of agriculture not popular throughout the country?

Answer

My uncle practices commercial farming which is a system of agriculture that involves cultivation of crops and livestock rearing for sale in the market. He mostly grows rice on the farm. Cash crops like sugarcane, tobacco, fibre crops and oilseeds are also grown.

The main features of commercial agriculture are:

1. Crops are grown mainly for sale.
2. Large farms are required.
3. Mechanised farming is practised.
4. It is prevalent in areas where farms are large and market economy is well developed.

5. It involves the use of High Yielding Variety (HYV) seeds, chemical fertilizers, insecticides and pesticides to obtain higher productivity.

Commercial agriculture is not popular throughout the country because in most other States there are small landholdings while commercial agriculture requires large farms.

Question 4

If you are given a choice, which type of agriculture you would like to practise and why?

Answer

I would like to practise organic farming because of the following reasons:

1. It protects the long term fertility of soils by maintaining organic matter levels, and creating optimised conditions for biological activity within the soil.
2. It provides crop nutrients indirectly using relatively insoluble nutrient sources which are made available to the plant by the action of soil micro-organisms.
3. It maintains nitrogen self-sufficiency through the use of legumes and biological nitrogen fixation as well as effective recycling of organic materials including crop residues and livestock manures.
4. It prevents weeds, diseases and pests by relying primarily on crop rotations, natural predator diversity, organic manuring, resistant varieties and limited thermal, biological and chemical intervention.
5. It provides attentive care and management to livestock, paying full regard to their evolutionary adaptations, behavioural needs and animal welfare issues with respect to nutrition, housing, health, breeding and rearing.

Question 5

The concept of organic food is fast catching up in metro cities. What are your views on this? Give reasons to support your answer. Are you ready to pay more for organic products?

Answer

I agree that the concept of organic food is fast catching up in metro cities. People are eager to switch to organic food because of the following reasons:

1. **Health Benefits** — Organic food is often perceived as healthier as it is produced without the use of synthetic pesticides, genetically modified organisms (GMOs), or artificial additives. It is believed to have higher nutritional value and lower levels of chemical residues.
2. **Environmental Impact** — Organic farming practices prioritize soil health, biodiversity, and the use of natural resources. It can help protect the environment, conserve water, and reduce pollution.

3. Support for Local Farmers — Organic food production often involves small-scale and local farmers who follow organic principles. Buying organic products can contribute to supporting local communities and sustainable agriculture practices.

Yes, keeping in mind my own health as well as the conservation of our environment, I would be amenable to pay more for organic products.